Stochastic Processes 2021/22 Semester 1

Extension Questions 3

These questions are an extension to Tutorial 3, intended for personal revision only. Answers are available for all questions, but worked solutions are only available for some.

1. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the state space $S = \{1, 2, 3, 4\}$ and has transition matrix

$$P = \begin{pmatrix} 1/3 & 0 & 2/3 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

   (a) i. Find the matrix giving the 2-step transition probabilities, and the matrix giving the 3-step transition probabilities.

   ii. If the chain is in state 3 at time $n = 71$, what is the probability that it is in state 1 at time $n = 74$?

   (b) Suppose that $\mathbf{P}(X_0 = i) = 1/3$ for $i = 1, 2, 3$ and $\mathbf{P}(X_0 = 4) = 0$.

   i. Find the distribution of the random variable X_3.

   ii. Find also

$$\mathbf{P}(X_0 = i \,|\, X_3 = 4), \qquad i = 1, 2, 3, 4$$

   i.e. find the initial distribution of the chain *given* that the chain is in state 4 after 3 steps.

2. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the state space $S = \{0, 1, 2\}$ and has transition matrix

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}.$$

   (a) Show by induction that the matrix of n-step transition probabilities is given by

$$P^n = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{pmatrix}.$$

   (b) Deduce that

$$\lim_{n\to\infty} p_{ij}^{(n)} = 1/3$$

   for all j, independently of the state i.

   (c) Suppose that $\mathbf{P}(X_0 = 0) = \mathbf{P}(X_0 = 1) = 1/2$, and $\mathbf{P}(X_0 = 2) = 0$

   i. Find an expression for $\mathbf{P}(X_n = i)$ for $i = 1, 2$ and all n.

   ii. Find also $\lim_{n\to\infty} \mathbf{P}(X_n = i)$ for $i = 1, 2$.

   (d) Suppose that $\mathbf{P}(X_0 = i) = 1/3$ for $i = 0, 1, 2$. Find $\mathbf{P}(X_n = i)$ for $i = 1, 2$ and all n.

3. An insurance company sets up a no claim discount scheme with 3 levels of discount (20%, 40% and 60%) and the base 'no discount' level. The rules by which the scheme operate are:

   - If the policy holder does not have an accident in a year they move up a class to a higher discount level or if in the highest level they stay in that level.
   - For every accident a policy holder has in a year they move done a discount level but can go no lower than 'no discount'. So for example if they are in the 40%, level and have 1 accident they move to the 20% level, and 2 or more to the no discount level.

   The number of accidents a policy holder experiences each year is Y, where for $i = 0, 1, 2, \ldots$

   $$\mathbf{P}(Y = i) = p^i(1 - p)$$

   for some parameter p.

   (a) When $p = 1/10$ find the probability that a policy holder who has the 20% discount has the 60% discount after 4 years.

   (b) For general p find an expression for the probability that a policy holder who has the 60% discount has no discount after 2 years.

4. For each of the Markov chains whose transition matrix is given below, determine the classes of intercommunicating states. Identify the closed classes and the vector of absorption probabilities associated with each of these. (In each case assume the states are labelled $1, 2, \ldots$.)

   (a)
   $$\begin{pmatrix} 1/4 & 0 & 0 & 0 & 0 & 0 & 3/4 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1/6 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3/4 & 0 & 1/4 & 0 \\ 1/2 & 0 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1/6 & 1/6 & 0 & 2/3 & 0 & 0 \end{pmatrix}$$

   (b)
   $$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2/3 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 1/3 & 2/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/6 & 0 & 1/2 \\ 1/6 & 0 & 0 & 5/6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/4 & 1/4 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/3 & 0 & 0 \end{pmatrix}$$

   (c)
   $$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 2/7 & 2/7 & 0 & 1/7 & 0 & 2/7 & 0 \\ 0 & 0 & 1/5 & 1/5 & 2/5 & 1/5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix}$$

5. A fair coin is tossed repeatedly with results $Y_0, Y_1, Y_2, \ldots$ that are 0 (for a tail) and 1 (for a head), with probability 1/2 for each.

   For $n \geq 1$ let $X_n = Y_n + Y_{n-1}$ be the number of heads in the $(n-1)$th and nth tosses. In other words, $X_n = 2$ if the $(n-1)$th and nth tosses were both heads, $X_n = 0$ if the $(n-1)$th and nth tosses were both tails etc.

   (a) Let $Z_n = (X_{n-1}, X_n)$. Is $\{Z_n\}_{n\geq 1}$ a Markov chain? If so write down its state space and transition matrix.

   (b) Consider instead the case where $X_n = Y_n \cdot Y_{n-1}$. In this case:

      i. Is $\{X_n\}_{n\geq 1}$ a Markov chain? If so write down its state space and transition matrix.

      ii. Is $\{Z_n\}_{n\geq 1}$ a Markov chain? If so write down its state space and transition matrix.

6. An absent-minded professor has 2 umbrellas that she uses when commuting from home to office and back. If it rains and an umbrella is available in her location, she takes it. If it is not raining, she always forgets to take an umbrella. Suppose that it rains with probability p each time she commutes, independently of other times. We want to understand whether she gets wet or not.

   (a) What is the possible state space for this problem? (Hint: Can you do it with only 3 states?)

   (b) Find the associated P matrix.

   (c) If the professor starts at home with both umbrellas, what is the probability she has at least one umbrella with her after 3 journeys?